

COURSE DESCRIPTION			
Program	Biomedical Engineering, Electronics and Communication, Electronics Engineering, Electronics Engineering (Embedded Systems), Electronics and Computer Engineering, Integrated Circuit Design & Fabrication, Micro Electronics		
Course Title	Electronic Devices and Applications		
Course Code	2042	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

This course introduces the fundamentals of AC to DC conversion and signal shaping techniques used in electronic systems. It explains the operation and characteristics of semiconductor devices, including bipolar and unipolar transistors. Overall, the course provides a strong foundation for analyzing and applying basic electronic devices in real-world electronic applications

Course Objectives

1	To analyze and apply the principles of AC to DC conversion and linear waveform shaping in basic electronic applications.
2	To explain and apply nonlinear waveform shaping and voltage multiplication concepts in electronics
3	To analyze the operation and characteristics of bipolar junction transistors in electronic circuits.
4	To understand the operation and characteristic behavior of field-effect and unijunction transistors in electronic circuits.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Passive components, semiconductor theory, diodes	1041	Elementary concepts of Electronics	I
Basic knowledge in Engineering Mathematics	1002	Fundamentals of Engineering Mathematics	I
Basic concepts of Electrical Mathematics	2001B	Mathematics for Electrical Sciences	II

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply AC to DC conversion and linear waveform shaping techniques in electronic systems.	Apply
CO2	Apply waveform shaping and voltage multiplication methods for signal conditioning	Apply
CO3	Explain the operation, configurations, and characteristics of bipolar junction transistors.	Understand
CO4	Explain the operation and characteristic behavior of field-effect transistors and unijunction transistors.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	2	3	2	1	-	-	-	-	1
CO2	3	2	2	3	2	1	-	-	-	-	1
CO3	3	2	1	2	1	-	-	-	-	-	1
CO4	3	2	1	2	1	-	-	-	-	-	1
Total	12	8	6	10	6	2	-	-	-	-	4
Correlation Level	3	2	1.5	2.5	1.5	1	-	-	-	-	1

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	

4	0	0	0	6	60	4	40	60	100	0	0	0	100
L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination													

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Linear wave shaping circuits	Half wave rectifier - Center tapped rectifier -Bridge rectifier - Compare rectifiers Filters - Application of filters in rectifiers RC differentiator -RC integrator	15	0
Module II	Nonlinear wave shaping circuits	Clipper circuits - Clamper circuits -Voltage Multipliers	15	0
Module III	Working principle of Bipolar Junction Transistor	Bipolar Junction Transistors - NPN Transistor- PNP Transistor Transistor biasing - Modes of operation Operation of NPN transistor Transistor configurations Input and output characteristics of CB Configuration and CE configuration	15	0
Module IV	Working principle of JFET, MOSFET and UJT	Junction Field Effect Transistors -N channel JFET MOSFET- Unijunction Transistor	15	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Half wave rectifier	Apply AC to DC conversion using half-wave rectification.	L	CO1	Apply	2
Center tapped rectifier	Apply full wave rectification using center tapped transformer	L	CO1	Apply	2
Bridge rectifier	Apply full wave rectification using bridge arrangement	L	CO1	Apply	2
Compare Rectifiers	Compare half wave, center tapped, and bridge rectifiers based on DC output voltage, ripple factor, PIV, and TUF	L	CO1	Understand	2
Filters	Explain the working of capacitor, inductance, and pi-filters used for ripple reduction in rectifier outputs	L	CO1	Understand	2
Application of filters in rectifiers	Explain the use of capacitor, inductance, and pi-filters in half-wave and full-wave rectifiers for ripple reduction	L	CO1	Understand	3
RC differentiator	Apply an RC differentiator to obtain differentiated output waveforms	L	CO1	Apply	1
RC integrator	Apply an RC integrator to obtain integrated output waveforms	L	CO1	Apply	1
Module II					
Clipper circuits	Apply a series diode positive clipper with and without bias to limit selected portions of an input waveform	L	CO2	Apply	2
Clipper circuits	Apply a parallel diode clipper with and without bias to limit selected portions of an input waveform	L	CO2	Apply	2
Clipper circuits	Apply a combinational diode clipper to limit both positive and negative portions of an input waveform	L	CO2	Apply	2
Clipper circuits	Apply positive, negative, and full-wave Zener diode clippers to limit the amplitude of input waveforms	L	CO2	Apply	2
Clamper circuits	Apply a positive clamper with and without bias	L	CO2	Apply	2
Clamper circuits	Apply a negative clamper with and without bias	L	CO2	Apply	2
Voltage multipliers	Apply halfwave and full wave voltage doubler to obtain twice the peak input voltage	L	CO2	Apply	2

Voltage multipliers	Apply a voltage tripler to obtain three times the peak input voltage	L	CO2	Apply	2
Module III					
Bipolar Junction Transistors	State the basic concept and function of a bipolar junction transistor	L	CO3	Understand	1
NPN Transistor	Explain the structure of NPN transistors.	L	CO3	Understand	1
PNP Transistor	Explain the structure of PNP transistors.	L	CO3	Understand	1
Operation of NPN transistor	Explain the operation of an NPN transistor.	L	CO3	Understand	2
Transistor configurations	Explain common transistor configurations (CB, CE, CC).	L	CO3	Understand	1
Input and output characteristics of CB configuration	Explain the input and output characteristics of the common-base configuration.	L	CO3	Understand	2
Input and output characteristics of CE configuration	Explain the input and output characteristics of the common-emitter configuration.	L	CO3	Understand	2
Modes of operation	Explain cutoff, active, and saturation modes of operation of CE and CB configurations	L	CO3	Understand	1
Load line	Explain load line, Q point,	L	CO3	Understand	1
Transistor biasing	Explain the need of transistor biasing.	L	CO3	Understand	1
Transistor biasing	Explain the methods of transistor biasing.- voltage divider bias, fixed bias	L	CO3	Understand	2
Module IV					
Junction Field Effect Transistors	State the basic concept and function of an FET	L	CO4	Understand	1
N channel JFET	Explain the principle of operation of an N-channel JFET.	L	CO4	Understand	2
N channel JFET	Explain the drain characteristics of an N-channel JFET.	L	CO4	Understand	2
MOSFET	Explain the principle of operation of an N-channel depletion MOSFET.	L	CO4	Understand	2
MOSFET	Explain the drain characteristics of an N-channel depletion MOSFET.	L	CO4	Understand	2
MOSFET	Explain the transfer characteristics of an N-channel depletion MOSFET.	L	CO4	Understand	1
Uni-Junction Transistor	Explain the structure of a unijunction transistor.	L	CO4	Understand	1
Uni-Junction Transistor	Explain the V-I characteristics of a Unijunction transistor.	L	CO4	Understand	2

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare a chart of household equipment that uses rectifier in it.	6
2	List the electronic equipment that works on DC supply	6
3	Prepare a flow chart showing the steps involved in converting AC to DC	6
4	Simulate RC integrator and differentiator using any simulation tool (like Tinkercad) Sketch and compare the output waveforms of RC differentiator and RC integrator.	6
Module II		
5	Tabulate the difference between biased and unbiased clipper circuits based on clipping level and application.	6
6	Draw and label positive and negative clamper circuits and indicate voltage shift direction	6
7	Using any simulation software study the changes in output waveforms of an RC integrator and differentiator with frequency of input waveforms.	6
8	List five real-life devices that use voltage multipliers	6
Module III		

9	Collect datasheets of any two BJTs and list their ratings (VCE, IC, gain, power).	6
10	Collect pictures of different transistor packages (TO-92, TO-220, metal can) and identify their pin configuration	6
11	By watching a relevant video from the internet, study the testing procedure for checking transistors. Prepare a document.	6
12	Prepare a short presentation (2-3 minutes) explaining the applications of transistors	6
Module IV		
13	Prepare a chart showing symbols, terminals, and basic structure of JFET, MOSFET, and UJT.	6
14	Collect datasheets of one JFET, one MOSFET, and one UJT and tabulate key parameters (IDSS, VGS(off), threshold voltage, ratings).	6
15	Prepare a short note on real-life applications of FETs and UJT	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	A Text Book of Applied Electronics	R S Sedha	S Chand
2	Basic Electronics and Linear Circuits	N N Bhargava, Kulshreshtha and S C Gupta	Tata McGraw Hill

REFERENCE			
Sl. No.	Title	Author	Publication
1	Electronic Devices and Circuits	Robert Boylestad	PHI
2	Electronic Devices and Circuits	Anil K Maini and Varsha Agarwal	Wiley India
3	Electronic Devices and Circuits	David A Bel	PHI

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.electronics-tutorials.ws	II & III
2	https://archive.nptel.ac.in/content/storage2/courses/117107095/lecturers/lecture_34/lecture34_page1.htm	IV
3	http://www.digimat.in/nptel/courses/video/122106025/L09.html	I
4	tinkercad.com	All
5	https://www.circuitlab.com/	All

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
------	--------	--------	-------	--------	-------	----------------------	-------	-------

2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60

SITTTR